# CLINICAL DATA MANAGEMENT PROTOCOL

*AI-driven Data Cleaning & Standardization — Screening Visit Dummy Dataset (CDM Training)*

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| Protocol Title: | Clinical data management protocol |
| Protocol No.: | CDM-TRN-W5-2026 |
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| Sponsor: | BIOTECHNIKA |
| Submitted by: | Suraj Anil Patil |
| Guided by: | Mrs. Nidhi Jha |

# 1. Introduction & Purpose

This protocol is prepared for Clinical Data Management (CDM) of the 20-subject dummy dataset that will be used as a part of the training course. This document reflects general information about the data's design, collection, specifies what data checks/edit checks should be performed on the collected data, and defines the standard of the deliverable data.

This paper aims to:

●describe the scope and general layout of the dummy dataset created for training;

●describe the list of variables to be collected, their format, and whether they are required or optional;

●define the specific validation (edit-check) process to identify discrepancies;

●outline the duties of the Clinical Data Manager (CDM) and query resolution procedure;

●report the quality issues discovered in the dummy dataset and how they were resolved;

●report the AI-assisted review of this protocol before submitting it for approval.

# 2. Synopsis

Study Design: Non-interventional training study with a single screening visit using a dummy dataset.

Purpose: To design and implement a clinical data management workflow (eCRF build, edit checks, query resolution, and dataset baseline summary) on a 20-subject dummy dataset and validate the process using AI-assisted review of this protocol.

Population: 20 subjects (3 sites), age 18 to 65 years old.

Endpoints: Completeness and accuracy of data collected at the Screening visit, including demographics, vital signs at screening, consent, and eligibility assessment; number and resolution of queries raised during edit-check validation.

Procedures: Collect the data for 20 subjects in Castor EDC and review it against the 20-item validation checklist, defining the data discrepancies, queries raised, and eligibility for each subject according to the informed consent process and inclusion/exclusion criteria.

# 3. Study Design

The study is observational/non-interventional (training dataset): there is no control or treatment group as no investigational product is involved. This is a training study; therefore, there is no blinding, and no randomization procedures are applied. The subject visit structure includes a single screening-only data collection. The study's duration is aligned with the week 5 of the CDM training course, where the data management process is designed for a single cycle of data entry and review.

# 4. Study Population

Inclusion Criteria

●A subject is entered into the system as a Site 1, 2, or 3 with an SCRxxx subject ID

●Subject age is between 18 and 65 years at the time of consent

●Informed consent is obtained (Consent Obtained = Yes) with a valid consent date.

Exclusion Criteria

●Vital signs are implausible (Height <140 cm, Height >210 cm, Weight <40 kg, and Weight >150 kg)

●A duplicate subject is identified within the site's subject entry (e.g., two subjects with the subject ID of SCR001)

Withdrawal Criteria

There are no subject withdrawal criteria as this is a screening-only dataset and no screening follow-up is planned.

## 5. Study & Dataset Overview

This dummy dataset simulates the clinical trial's screening visit and includes the records for 20 subjects across 3 sites. For each subject, demographical data, vital signs, consent, and eligibility were entered.

| Metric | Value |
|---|---|
| Total subjects (records) in dataset | 20 |
| Total sites represented | 3 (Site 1, Site 2, Site 3) |
| Visit represented | Screening (single visit) |
| Subjects eligible | 19 |
| Subjects not eligible | 1 (SCR005 – missing Consent Date) |
| Consent obtained = Yes | 20 of 20 |
| Inclusion criteria met = Yes | 20 of 20 |
| Exclusion criteria met = Yes (expected: 0) | 0 |

*Note: SCR005 is recorded as not eligible because the Consent Date field is missing despite Consent Obtained = Yes; this is treated as a protocol deviation pending query resolution, not a true screen failure.*

## 6. Roles & Responsibilities

| Role | Responsibility |
|---|---|
| Clinical Data Manager (CDM) | Designs eCRF/database build, writes and executes validation checks, reviews queries, ensures dataset integrity, prepares this protocol and the baseline summary. |
| Principal Investigator (PI) / Site Staff | Records source data, completes eCRF entries in Castor EDC, responds to data queries raised by the CDM. |
| Clinical Research Associate (CRA) / Monitor | Performs source data verification, confirms consent and eligibility documentation, escalates discrepancies. |
| Course Guide (Mrs. Nidhi Jha) | Reviews and approves the protocol, validation plan, and training deliverables submitted for this course. |
| Student / Trainee (Data Management) | Builds the dummy dataset review, drafts the protocol, validation rules and baseline summary as part of weekly CDM training tasks. |

## 7. Data Collection Instrument (Castor EDC)

Data are captured electronically in Castor EDC using a Screening/Baseline eCRF, structured as follows:

- Demographics – Subject ID, Site ID, Initials, DOB, Age, Sex.
- Baseline / Eligibility – Screening Date, Consent Obtained, Consent Date, Inclusion Criteria Met, Exclusion Criteria Met, Eligible.
- Vital Signs – Height (cm), Weight (kg), BMI (derived).

Field level checks (range, mandatory, derivation) are embedded in Castor EDC where possible, otherwise checks are performed during data entry.

# 8. Dataset Baseline Summary

## 8.1 Number of Subjects

The dataset consists of 20 subjects (SCR001-SCR017, SCR019, SCR020, and SCR034), at 3 sites, and at screening visit (see Section 5 for details). The table below presents the baseline summary sheet of the dataset: an overview of the number of subjects, the variables, their expected value type/format and whether they are mandatory or not.

## 8.2 Key Variables, Expected Formats & Mandatory/Optional Status

| Variable Name | Description | Data Type | Expected Format / Allowed Values | Mandatory / Optional |
|---|---|---|---|---|
| SUBJECT ID | Unique subject identifier | Text | SCRxxx (3-digit, zero-padded, sequential) | Mandatory |
| SITE ID | Site enrolling the subject | Numeric | Integer, per approved site list | Mandatory |
| INITIALS | Subject initials | Text | 2 letters, uppercase | Mandatory |
| VISIT NAME | Name of the visit | Text | Fixed value: "Screening" | Mandatory |
| SCREENING DATE | Date subject was screened | Date | DD-MM-YYYY | Mandatory |
| DOB | Date of birth | Date | DD-MM-YYYY | Mandatory |
| AGE YRS | Age in completed years | Numeric | Whole number; plausible adult range (18-65, per protocol) | Mandatory |
| SEX | Biological sex | Categorical | Male / Female | Mandatory |
| HEIGHT CM | Height in centimetres | Numeric | Plausible range approx. 140-210 cm | Mandatory |
| WEIGHT KG | Weight in kilograms | Numeric | Plausible range approx. 40-150 kg | Mandatory |
| BMI | Body Mass Index | Numeric | Derived: Weight / $(\text{Height}/100)^2$ | Mandatory (derived) |
| CONSENT OBTAINED | Informed consent status | Categorical | Yes / No | Mandatory |
| CONSENT DATE | Date informed consent signed | Date | DD-MM-YYYY | Conditional* |
| INCL CRITERIA MET | Inclusion criteria status | Categorical | Yes / No | Mandatory |
| EXCL CRITERIA MET | Exclusion criteria status | Categorical | Yes / No | Mandatory |
| ELIGIBLE | Overall eligibility outcome | Categorical | Yes / No | Mandatory (derived) |

**Conditional: mandatory only when a stated condition is met (Consent Date is required once Consent Obtained = Yes).*

## 8.3 Mandatory vs Optional – Summary

All 16 variables in this dataset are either Mandatory or Conditional; there are no optional items in the Screening/Baseline. This is indicated by the requirement that the subject is only eligible if all items are completed correctly.

# 9. Data Quality Review & Query Management Process

The following describes the data review process that will occur once the data has been entered into the system:

- Edit checks defined in Section 9 are run through the entire dataset after each data entry cycle.

- Any record that fails an edit check will be flagged and a query will be raised to the site via Castor EDC.
- Sites are expected to respond to queries within 5 working days and the CDM will review and close queries as they are responded to.
- Queries that are not resolved will be escalated to the CRA/PI and entered into the query tracker until the database is locked.
- A final check of all fields will happen before the database is locked to ensure there are no outstanding queries or issues with the data before analyses commence.

## 10. Data Quality Observations Identified in This Dataset

The following issues were identified in the current 20-subject dummy dataset during baseline review and should be raised as queries:

| ID | Subject | Concern |
|---|---|---|
| EC-02 | SCR016 | Invalid date: 32-07-2026 – not a valid calendar date |
| EC-04 | SCR011 | Height = 300 cm – highly unlikely for an adult – likely a transcription error |
| EC-04 | SCR012 | Height = 125 cm – unlikely for an adult |
| EC-05 | SCR020 | Missing weight – a mandatory field |
| EC-08 | SCR005 | No value for CONSENT DATE even though CONSENT OBTAINED = YES – ineligible subject (pending query resolution) |
| EC-01 | SCR018, SCR034 | Site has 20 subjects but SCR018 is missing, and SCR034 appears instead of SCR018 |

# 11. Screenshot of the Original Messy Dataset in Castor

This section describes where the data quality issues from Section 10 can be found within Castor EDC – both as paper documentation and implemented computerized checks at the data entry forms and in the Validations dashboard. The four panels below have been merged into one image for easier viewing (Panels A-D).

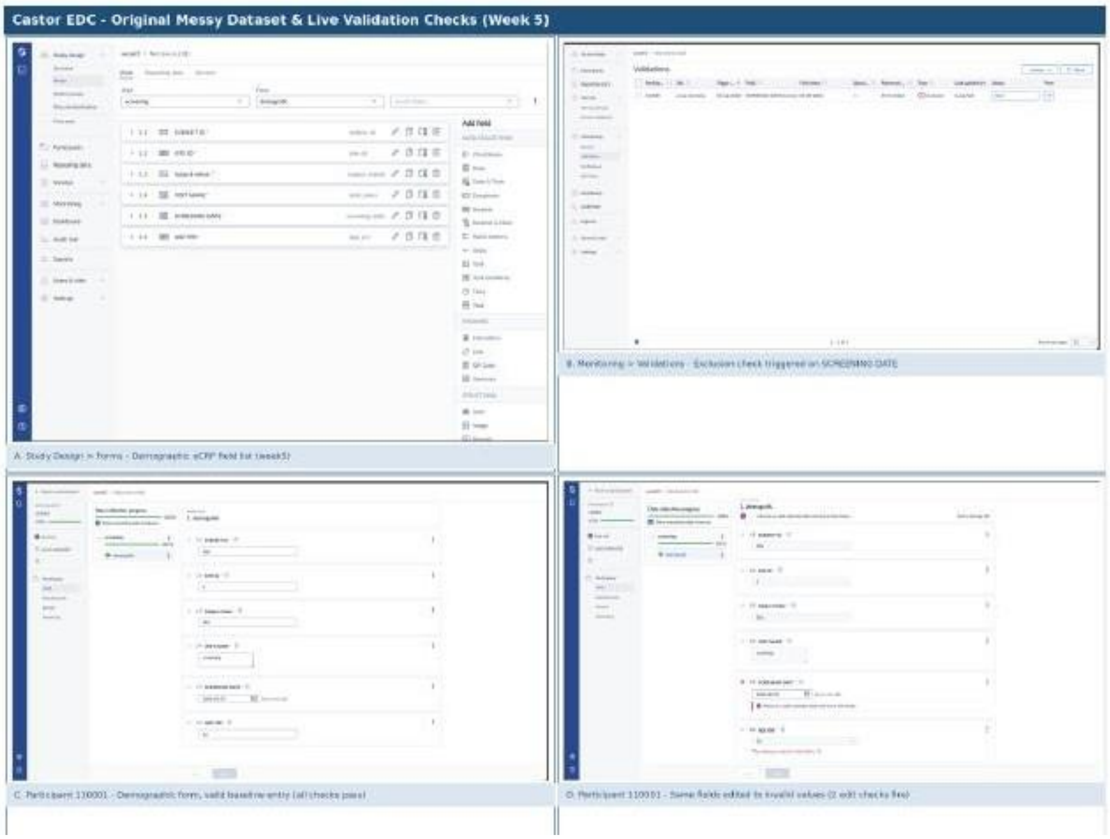


*Figure 1: Castor EDC – eCRF build (Panel A), Validations dashboard (Panel B), and a participant record shown in both a clean state (Panel C) and an edit-check failure state (Panel D).*

## Panel A – Study Design > Forms (Demographic eCRF)

This panel represents the field list for the Demographic form within the "screening" visit for this Week 5 study. It was populated with the following fields in exact correspondence with Section 8.2 of this protocol: SUBJECT ID, SITE ID, Subject initials, VISIT NAME, SCREENING DATE, and AGE YRS, as a confirmation that this eCRF was built according to the baseline summary rather than diverging from it.

## Panel B – Monitoring > Validations (system-raised query)

This panel contains the Castor-generated edit-check issue raised as a query record, and not as a note record within a document. For Participant 110001 at the indicated site "Surya Memorial", an Exclusion-type validation was triggered by the Screening Date value entry. The operator ">=" and the reference date "05-07-2026" were used for screening against the entered screening date "02-08-2026". This query currently has a status of "New", meaning it is awaiting review according to the Section 9 query-management workflow. It corresponds to edit check EC-02 Screening Date – must be a valid date and not outside the expected range.

## Panel C – Participant 110001, valid baseline entry

This panel contains the same participant's valid baseline entry for reference. It corresponds to the same Demographic form as per Panel A but was populated with a valid entry according to the baseline protocol. Subject ID "006", Site ID "2", Subject Initials "001", "screening" visit name, Screening Date "2026-05-07", Age

Yrs "32" all passed the validation edit checks with no queries raised. All fields are marked green and the form is completed at 100%. It serves as a "before" reference example for a valid baseline entry.

### Panel D – Same participant, edit checks firing on invalid values

Same participant's baseline entry but with altered values for the fields Screening Date and Age Yrs to demonstrate the "after" invalid entry with edit checks firing in real time rather than upon the end of the whole eCRF entry session. Screening Date "2026-08-02" is marked red with an edit check stating "Must be a valid calendar date and not in the future" – which corresponds to EC-02 Screening Date. Age Yrs entry of "30" is invalid because "The minimum value for this field is 31" – consistent with EC-03/04 age range plausibility edit checks. In this instance, Castor enforces field minimums rather than the data entry personnel entering invalid values.

## 12. AI-Assisted Protocol Review Summary

As a part of this training course, this protocol was subjected to AI-assisted review using AI model assistance (ChatGPT, Gemini, Claude) to improve its clarity, completeness, and consistency prior to submitting it to the course guide for review. The main areas of improvement fall under these categories:

### 12.1 Clarity

- Field descriptions were made more descriptive rather than referring to the raw dataset, where applicable. For example, the screening date is described as Screening Date rather than visit date.
- Terminology was standardized across the document for improved consistency.

### 12.2 Completeness

- A formal Roles & Responsibilities section was added, which was implicitly contained within the stepwise submission instructions but was not explicitly stated elsewhere in this protocol.
- A formalized Data Validation Plan was added along with the corresponding edit checks, fields, and actions.
- The Query Management workflow section was added to describe how query records are managed within Castor EDC.

### 12.3 Consistency

- The fields identified as Mandatory/Required were cross-referenced with the relevant edit check fields and actions to ensure consistency.
- The data quality observations listed in Section 10 were matched to their corresponding edit check identifiers as per Section 9.
- The baseline entry summary count from Section 5 was matched to the inclusion/exclusion criteria logic from Sections 9/10 to ensure consistency, such as the SCR005 count being indicative of an ineligible participant.

## 13. References


- Castor EDC — Electronic Data Capture Platform, https://www.castoredc.com
- ICH E6(R2) Good Clinical Practice Guideline — data management and quality principles referenced for protocol structure

- CDM training course materials, guided by Mrs. Nidhi Jha
- AI-assisted review tools used: ChatGPT, Gemini, Claude.